Supplementary Information for: Hard thermodynamic constraints and the temperature extrapolation of learned water isotherms in metal-organic and covalent organic frameworks: a pre-registered test

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Supplementary Table S1: Data

Material	Class	Set	Data	Source DOI	T range (K)	n_T	Points per T	p/p_{sat} range	q_{ref} (mol kg^{-1})
COF-432	COF	conf.	vector	10.1021/jacs.9b13094	283.15–313.15	3	28–29	0.063–0.870	16.04
COF-482-hydrazide	COF	conf.	raster	10.1021/acscentsci.2c00398	288.15–308.15	3	25–51	0.08–0.898	34.70
COF-SO_3H	COF	conf.	vector	10.1002/cssc.202301906	298.15–318.15	3	18–18	0.0003–0.900	17.04
HFPTP-DMePDA-COF	COF	conf.	raster	10.1038/s41467-021-27128-4	283.15–298.15	3	9–18	0.39–0.897	7.91
HFPTP-PDA-COF	COF	conf.	raster	10.1038/s41467-021-27128-4	283.15–298.15	3	24–32	0.0041–0.898	16.78
NO-PI-3-COF	COF	conf.	vector	10.1021/jacs.3c02572	288.15–308.15	3	30–37	0.0091–0.882	12.74
PI-3-COF	COF	conf.	vector	10.1021/jacs.3c02572	288.15–308.15	3	28–37	0.01–0.887	24.80
TFBCz-PDA-COF	COF	conf.	raster	10.1038/s41467-021-27128-4	283.15–298.15	3	29–30	0.0023–0.899	27.26
Al-fumarate	MOF	conf.	raster	10.1021/acscentsci.9b00745	283.15–323.15	3	56–59	0.049–0.892	23.86
CALF-25	MOF	conf.	ISODB	10.1021/ja306812r	298.00–313.00	3	17–20	0.018–0.894	3.75
Co_2Cl_2(BTDD)	MOF	conf.	vector	10.1021/acscentsci.7b00186	283.00–298.00	3	31–34	0.0072–0.862	53.97
Mg-SU-102	MOF	conf.	numeric	10.1038/s41467-025-59551-2	288.15–328.15	9	21–21	0.0013–0.819	21.11
Zn_2(bptc)	MOF	conf.	ISODB	10.1021/ja060946u	288.00–318.00	4	25–33	0.023–0.885	8.38
TFPPy-PDA-COF	COF	dev.	raster	10.1038/s41467-021-27128-4	283.15–298.15	3	7–15	0.46–0.900	35.41
CuBTC	MOF	dev.	ISODB	10.1016/j.ijheatmasstransfer.2012.07.068	293.00–358.00	9	23–41	0.0029–0.508	25.29
MIL-100	MOF	dev.	ISODB	10.1016/j.ijheatmasstransfer.2012.07.068	293.00–358.00	9	25–35	0.011–0.603	18.11
MIL-160	MOF	dev.	ISODB	10.1002/adma.201502418	293.00–313.00	3	32–54	0.0052–0.884	21.79

Supplementary Table S1. The 13 confirmatory materials (conf.) and the four development materials (dev.) after the inclusion rules and the preprocessing. Data: how the points were obtained (vector or raster digitisation of the published figure, numeric source data, or the NIST adsorption database, ISODB). n_T: number of temperatures; points per temperature with $p/p_{\mathrm{sat}} \leq 0.90$; q_{ref}: the largest uptake over all temperatures in this range, used only for scoring. Every decision and its reason is in data/inclusion_log.csv.

Supplementary Tables S2 to S4: Confirmatory results in detail

			$R_m = E_P/E_X$						
Material	Class	n_{train}	H1b V-pen	H1c A-hard	H1d F-none	H1e C1	H1a F-hard	V-none	F-rel
COF-432	COF	2	0.86	0.83	0.37	0.98	0.70	0.84	1.58
COF-482-hydrazide	COF	2	0.95	1.18	0.35	1.03	0.22	0.96	0.72
COF-SO_3H	COF	2	10.91	9.92	6.19	8.58	6.04	9.80	5.35
HFPTP-DMePDA-COF	COF	2	1.23	4.60	0.51	0.74	1.14	0.94	0.95
HFPTP-PDA-COF	COF	2	0.99	2.28	1.01	1.01	0.85	1.00	0.95
NO-PI-3-COF	COF	2	2.47	1.55	1.07	2.47	0.20	2.09	2.69
PI-3-COF	COF	2	1.10	4.81	1.50	2.20	0.22	1.45	1.99
TFBCz-PDA-COF	COF	2	0.78	3.14	0.99	0.97	0.39	0.94	0.83
Al-fumarate	MOF	2	0.87	0.49	0.28	0.90	0.39	0.94	1.19
CALF-25	MOF	2	0.94	0.98	0.39	0.59	0.77	0.90	0.41
Co_2Cl_2(BTDD)	MOF	2	1.44	1.13	1.02	0.92	0.59	1.38	1.92
Mg-SU-102	MOF	8	1.01	0.73	0.24	1.56	0.28	1.07	0.31
Zn_2(bptc)	MOF	3	1.27	0.32	0.66	1.66	0.49	1.34	0.67
Hodges-Lehmann R			1.10	1.51	0.63	1.23	0.49	1.14	1.13

Supplementary Table S2. Error ratio of P against each comparator per material, $R_m = 10^{d_m}$, with d_m the difference of the mean of $\log_{10} E$ over E-high and E-low; below one, P extrapolated better. n_{train}: the smallest number of training temperatures over the two tasks. The last row is the Hodges-Lehmann estimate over the 13 materials (main text, Fig. 2a). H1a, V-none and F-rel are descriptive comparisons.

Model	Fits	In domain	Wilson 95 %	Band	Strict	All tasks	Class
P	26	0	0.00–0.13	0	0	0/39	none
F-hard	26	0	0.00–0.13	0	0	0/39	none
A-hard	26	0	0.00–0.13	0	0	0/39	none
C1	26	0	0.00–0.13	0	0	0/39	none
P-K	2	0	0.00–0.66	0	0	0/3	none
V-pen	26	9	0.19–0.54	10	11	12/39	some
V-none	26	8	0.17–0.50	9	10	12/39	some
F-pen	26	13	0.32–0.68	12	24	19/39	often
A-pen	26	16	0.43–0.78	16	21	25/39	often
A-none	26	15	0.39–0.74	15	19	24/39	often
F-rel	26	22	0.66–0.94	22	26	34/39	often
F-none	26	26	0.87–1.00	26	26	39/39	often

Supplementary Table S3. Violating fits per model over the 26 fits of E-high and E-low. In domain: the pre-registered count, a fall of more than 2 % of q_{ref} below the running maximum or an uptake below -2 % of q_{ref}, inside the penalty domain; Wilson 95 % interval of its rate. Band: the same rule applied to the rest of the grid, up to 20 K beyond the measured temperatures and to 0.95 p_{sat}. Strict: any decrease of uptake with pressure or any negative uptake anywhere on the grid, with no tolerance. All tasks: the in-domain count adding I-mid. Class: the pre-registered word.

Model	Fits	Median range	Largest range	Range above 1	Selected best	Selected worst
P	26	0.04	2.34	1	6	9
V-pen	26	0.11	1.46	1	5	5
V-none	26	0.13	1.76	2	10	7
A-hard	26	0.01	0.28	0	12	5
A-pen	26	0.03	0.54	0	7	4
A-none	26	0.04	1.41	2	6	4
C1	26	0.01	2.84	1	4	2
F-hard	26	0.54	1.48	5	7	7
F-pen	26	0.67	1.28	3	4	2
F-none	26	0.26	0.95	0	8	5
F-rel	26	0.18	0.97	0	3	10
P-K	2	0.34	0.47	0	0	1

Supplementary Table S4. Seed spread of the held-out error over the E-high and E-low fits. Range: the difference between the largest and the smallest $\log_{10} E$ over the 5 seeds of one fit, in decades; Range above 1: fits whose seeds differ by more than a factor of ten. Selected best (worst): fits in which the seed with the lowest training loss, the one reported, had the best (worst) held-out error of the five.

Supplementary Note 1: Can each verdict fail? Synthetic worlds and power

Before the freeze, the whole pipeline (the tuned configurations, fitting from 5 seeds, the violation counts and the analysis code) was run in synthetic worlds that share no uptake data with the confirmatory set. Each world has 13 materials; material i uses the temperatures of confirmatory material i. An expected outcome was written for each world before it ran. In KS0 the truth is a two-site Sips isotherm with van't Hoff affinities, inside the family of P; in DA it is a Dubinin-Astakhov curve, inside the family of the hard characteristic curve (A-hard); in FW it is a Kirchhoff-Sips isotherm with a large heat capacity, so that van't Hoff is wrong; FLAT has one weakly curved Langmuir site; SUB has a sub-linear Sips isotherm outside the family of P and inside that of the penalty model V-pen.

The comparison of P with A-hard discriminated in both directions: P extrapolated better in KS0 ($R = 0.52$, exact 95 % interval 0.35 to 0.70) and A-hard better in DA ($R = 1.26$, 1.07 to 1.49). Three expectations were missed. In FW the comparison of P with the free temperature network F-hard still read "P better" ($R = 0.33$): with two training temperatures the free network cannot extrapolate, so this comparison cannot fail and is reported descriptively. In FLAT the unconstrained network F-none violated physics "often". In SUB, built for V-pen to win, P was slightly better ($R = 0.96$, 0.94 to 0.99); the mixture of Langmuir terms in P approximates a sub-linear isotherm over the measured range. The accuracy comparison of hard constraints with penalties therefore could not be made to fail synthetically, and it is tested as non-inferiority.

The power of each verdict was simulated with the frozen analysis code at $n = 13$ (2000 simulations per setting). At the spread expected for the comparison with V-pen (0.158 decades), the non-inferiority criterion is met with probability 0.0225 when the true ratio is at the margin, $R = 1.25$, and 0.51 (normal) or 0.74 (Student t, 3 degrees of freedom) when the two models are equally accurate. A "not met" outcome is therefore likely even under equal accuracy and is never read as evidence that the constraints cost accuracy.

Supplementary Note 2: Synthetic truths for the isosteric heat

The heat test uses five truth families with exact heats. Their shape parameters are fitted by bounded least squares to each confirmatory material's middle-temperature isotherm; their thermodynamic parameters are fixed in advance: KS, a Langmuir site with $Q_0 = 55$ kJ mol^{-1} plus a cooperative Sips site with $Q_0 = 48$ kJ mol^{-1}, a heat capacity of 40 J mol^{-1} K^{-1} and an exponent of at least 1.5, inside the family of the Kirchhoff variant of P and near that of P; DA, a Dubinin-Astakhov curve $q = W_0 \exp[-(A/E)^n]$ with $1 \le n \le 6$, inside the family of the characteristic curve; FG, a Fowler-Guggenheim isotherm with $Q_0 = 45$ kJ mol^{-1} and a lateral attraction between 0 and $3.9\,RT_{\min}$, inside neither; HS, strong hydrophilic sites, a Langmuir site with $Q = 80$ kJ mol^{-1} holding at most 0.3 q_{ref} plus a Sips site with $Q = 46$ kJ mol^{-1}, inside the family of P with a heat that falls steeply with loading; TC, a Sips isotherm with $Q = 50$ kJ mol^{-1} whose capacity falls by 0.2 % per kelvin, inside the family of P and outside that of the characteristic curve. Gaussian noise of 0.4 % of q_{ref} is added with a fixed seed per material and family.

Supplementary Note 3: Digitisation error

The digitiser was tested against published figures whose numeric data are also published, before any model was fitted (logs/digitisation_error.json, logs/digitisation_validation.json). On the vector figure of Mg-SU-102 the extraction from the PDF drawing commands recovered 198 of 198 points with a root-mean-square uptake error of 7×10^{-6} g g^{-1} and a largest error of 5×10^{-5} g g^{-1}. The same figure rasterised at 150, 300 and 600 dpi and read by template

matching gave root-mean-square uptake errors of 6.3×10^{-4}, 4.0×10^{-4} and 3.5×10^{-4} $\mathrm{g\,g^{-1}}$, with no false detection; markers drawn hollow are not read by the raster method. On raster figures of HKUST-1 from another paper, the root-mean-square uptake error per series was 0.11 to 0.19 % of the axis span, or 0.0012 to 0.0034 of the series' largest uptake, again with no false detection; markers hidden under other markers are missed (recall of the visible markers 0.91 to 1.00). On vector figures of two further papers read through their legend colours, every point was recovered and the root-mean-square uptake error was 0.04 to 0.08 % of the axis span. The noise level of the synthetic tests, 0.4 % of q_{ref}, is the largest raster error per isotherm maximum rounded up.